

A Unified Machine Learning Framework and Efficient Pattern Mining for Dynamic Price prediction in E-Commerce

¹Sumathi T., ²Ashwin K., ²Priyadharshini R., ²Rajeshwari L.

¹Faculty, Department of Information Technology

²Students, Department of Information Technology

Dr. Mahalingam College of Engineering and Technology (Autonomous), Pollachi, Tamil Nadu, India

Emails: sumathi.sae@gmail.com, achusuchu123@gmail.com, priyaruby2005@gmail.com, raji1442006@gmail.com

Abstract—Dynamic pricing is essential for modern e-commerce platforms, allowing retailers to adjust product prices in response to market fluctuations, consumer preferences, competitor actions, and seasonal variations. Developing a comprehensive dynamic pricing system involves integrating multiple computational components, such as scalable data extraction, time-series preprocessing, pattern discovery, predictive modeling, elasticity analysis, and reinforcement learning-based decision-making.

This paper introduces a unified framework that merges machine learning, pattern mining, and reinforcement learning to enable real-time dynamic pricing in e-commerce settings. The system features: (i) an automated Selenium-based pipeline for daily price scraping, (ii) a reliable ETL workflow with data normalization and SQLite time-series storage, (iii) pattern detection based on volatility and change-point methods for analyzing product behavior, (iv) a CatBoost regression model tailored by category for forecasting next-day prices, (v) a hybrid elasticity estimation module employing simulated Bayesian priors to assess demand sensitivity, and (vi) a Thompson Sampling multi-armed bandit algorithm for selecting optimal prices under uncertainty.

Conducted experiments on a dataset of 1,246 products tracked over multiple days demonstrate forecasting errors ranging from 1.56% to 8.70% MAPE across categories, with the reinforcement learning agent showing effective revenue-focused convergence. Results confirm that combining elasticity insights with reinforcement learning enhances pricing strategy performance compared to static or purely machine learning approaches. The proposed architecture provides a scalable, production-ready solution for dynamic pricing in real-world online retail applications.

Index Terms—Dynamic pricing, machine learning, time-series forecasting, pattern mining, change-point detection, price elasticity, reinforcement learning, multi-armed bandits, Thompson Sampling, CatBoost, e-commerce analytics.

I. INTRODUCTION

The expansion of e-commerce platforms has fundamentally changed how consumers engage with online markets. With countless products competing for attention, dynamic pricing has become a critical strategy for enhancing profitability, managing stock levels, and adapting to real-time market signals. Unlike fixed pricing, dynamic pricing adjusts product prices continuously in response to demand shifts, competitor pricing, seasonal patterns, user interactions, and product-specific trends.

Leading e-commerce companies like Amazon and Flipkart employ sophisticated pricing systems that leverage machine learning, statistical analysis, and reinforcement learning to recommend optimal prices. Building an integrated framework for this purpose, however, involves several challenges: ensuring reliable real-time data collection, processing high-quality time-series data, implementing robust forecasting methods, providing interpretable pattern discovery, and designing scalable decision algorithms capable of adapting to uncertain and changing environments.

Most prior studies focus on isolated aspects such as price forecasting, demand elasticity, or reinforcement learning independently. Only a few attempt to combine these components into a comprehensive pipeline covering the full pricing workflow—from data acquisition and prediction to optimization. Furthermore, many assume the presence of extensive historical sales or demand information, which is often missing for newer or less prominent products.

This paper aims to fill these gaps by presenting a unified dynamic pricing architecture that integrates: (i) automated web scraping with ETL pipelines to build accurate time-series datasets, (ii) pattern mining methods including volatility measurement and change-point analysis to capture product behavior, (iii) a CatBoost-based machine learning model for category-specific next-day price forecasts, (iv) a hybrid elasticity estimation technique using simulated Bayesian priors for demand response approximation, and (v) a Thompson Sampling reinforcement learning agent to optimize price selection for revenue maximization.

We implement and validate this system using actual e-commerce product data gathered over multiple days. Experimental results show strong prediction accuracy, reliable behavior pattern recognition, and efficient convergence of the bandit agent toward profitable pricing decisions. This integrated solution offers a scalable, production-ready platform for real-time dynamic pricing in contemporary online retail environments.

II. RELATED WORK

Dynamic pricing has been extensively studied in various fields such as e-commerce, retail, transportation, and revenue management. Existing research typically falls into several categories, including traditional econometric models, machine learning approaches, pattern-based analysis, elasticity estimation, and reinforcement learning-driven price optimization.

A. Traditional and Econometric Pricing Models

Traditional dynamic pricing approaches typically rely on demand curves estimated through linear or log-linear regression models, inventory-based frameworks, or established revenue management principles [1]. These methods presuppose abundant historical sales records and stable demand patterns, conditions that rarely apply to modern e-commerce settings marked by short product lifecycles and rapid price fluctuations.

B. Machine Learning for Price Forecasting

Recent studies employ machine learning approaches including Random Forest, Gradient Boosting Machines, and neural networks for retail price and demand forecasting [2]. Time-series models such as ARIMA, Prophet, and LSTM have been utilized to capture seasonal trends and short-term fluctuations in e-commerce pricing data [3]. However, these techniques often fail to account for critical product-specific indicators like volatility patterns, market saturation, and abrupt structural changes that significantly affect pricing decisions.

C. Pattern Mining and Behavioral Analysis

Methods for extracting price patterns primarily target volatility detection, seasonal cycles, and anomalous price shifts. Change-point detection techniques such as BOCPD and Ruptures effectively identify sudden market regime changes [4]. However, these valuable approaches are seldom embedded directly into forecasting pipelines or pricing optimization frameworks to enhance decision quality.

D. Price Elasticity Estimation

Price elasticity estimation forms the foundation of economics and revenue management, quantifying consumer responsiveness to price variations. Conventional techniques employ regression analysis on historical sales records [5]. However, emerging or niche products frequently lack adequate sales data, complicating reliable elasticity estimation. Advanced Bayesian approaches with simulated priors have been developed for data-limited scenarios [6], yet their integration into operational e-commerce pricing platforms remains uncommon.

E. Reinforcement Learning for Dynamic Pricing

Reinforcement learning has gained traction for dynamic pricing challenges, allowing agents to discover optimal pricing strategies through reward feedback. Multi-armed bandit methods including UCB and Thompson Sampling have proven effective across online retail, ride-sharing services, and advertising platforms [7]. Yet most RL studies presuppose established

demand curves or substantial transaction histories—conditions frequently unmet for numerous e-commerce products.

F. Research Gap Summary

While there is extensive work on forecasting, elasticity estimation, and RL approaches separately, few studies present a comprehensive architecture that unifies these components into a single, scalable, and production-ready system. In particular, integrating (i) automated real-time time-series data collection, (ii) pattern mining for behavioral insights, (iii) machine learning forecasting, (iv) elasticity estimation for cold-start scenarios, and (v) reinforcement learning for revenue maximization remains an open challenge. This paper proposes such a unified framework designed specifically for the demands of e-commerce dynamic pricing.

III. SYSTEM ARCHITECTURE

The proposed framework integrates multiple analytical components into a cohesive dynamic pricing pipeline designed to enable real-time pricing decisions for e-commerce platforms. The architecture comprises seven core modules: (i) automated data collection, (ii) ETL processing with database storage, (iii) pattern mining and behavioral categorization, (iv) change-point detection, (v) machine learning-based price forecasting, (vi) elasticity estimation using Bayesian simulated priors, and (vii) reinforcement learning for price optimization. Figure 1 illustrates the high-level system design.

A. Overall Workflow

The end-to-end process operates as follows:

- 1) **Daily scraping:** A Selenium-powered crawler retrieves product prices, ratings, metadata, and availability from e-commerce websites.
- 2) **ETL preprocessing:** Collected raw data is cleaned, validated, and normalized before storing in a SQLite time-series database.
- 3) **Pattern mining:** Computation of statistical features such as volatility, trend slope, distinct price points, and price range, followed by classification of product behavior into categories like stable, volatile, downward-trending, or suspicious.
- 4) **Change-point detection (CPD):** Identification of abrupt price changes using the PELT algorithm via the Ruptures library.
- 5) **ML forecasting:** Next-day price predictions are made using category-specific CatBoost models that leverage lag features, moving averages (MA3, MA7), volatility indicators, and attributes derived from change-point detection.
- 6) **Elasticity estimation:** When sales data is unavailable, Bayesian simulated priors based on volatility metrics serve as proxies to estimate demand elasticity and define optimal price ranges.
- 7) **Bandit pricing agent:** Thompson Sampling and UCB1 algorithms explore a set of candidate prices to identify the revenue-maximizing price under uncertainty.

B. Modular Design

Each component is encapsulated as an independent executable script, facilitating scalable deployment and flexible integration:

- **Scraper module** (`amazon_scraper_selenium.py`)
Automates price extraction from product listings.
- **ETL and DB module** (`etl_clean_store.py`)
Performs data cleaning and updates the SQLite time-series storage.
- **Pattern mining module** (`pattern_classification.py`)
Produces behavioral metrics and classifies product price trends.
- **CPD module** (`change_point_engine.py`)
Detects sudden changes in pricing dynamics.
- **ML model trainer** (`train_price_model.py`)
Trains category-wise CatBoost models and evaluates feature importances.
- **Elasticity estimator** (`elasticity_engine.py`)
Calculates elasticity priors and optimal price multipliers from volatility.
- **Bandit pricing engine** (`bandit_pricing_agent.py`)
Learns the optimal price strategy through reward-based exploration.

C. Integration Layer

A central pipeline orchestrator (`run_daily_pipeline.py`) coordinates the execution of all modules, enabling automated daily updates, continuous data integration, and ongoing price optimization. The system is designed for straightforward deployment within pricing dashboards and APIs to support real-time e-commerce pricing workflows.

IV. METHODOLOGY

This section describes the fundamental analytical components within the proposed dynamic pricing system. The framework integrates modules for data gathering, preprocessing, pattern extraction, machine learning forecasting, elasticity inference, and reinforcement learning. Each module functions independently but contributes to a cohesive pricing pipeline.

A. Data Collection via Selenium Scraping

Product price data was gathered daily from online retailers using a Selenium-driven browser automation script. The scraper captures essential product attributes such as price, rating, number of reviews, availability status, and image URL. To ensure ethical and compliant scraping, techniques like rotating user-agent strings and randomized delays between requests were employed. Each run appends new records to the existing historical dataset, building a longitudinal time series:

- **Input:** A list of product URLs maintained in a watchlist.
- **Output:** Raw price log files appended to `amazon_price_history.csv`.

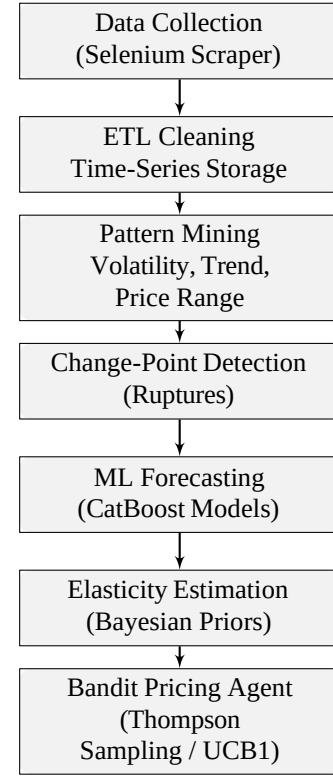


Fig. 1. Unified system architecture integrating data acquisition, preprocessing, pattern mining, forecasting, elasticity modeling, and reinforcement learning.

For a given product i , the daily scraped record at time t is expressed as:

$$x_{i,t} = \{\text{price}_{i,t}, \text{rating}_{i,t}, \text{reviews}_{i,t}, \text{availability}_{i,t}\}.$$

B. ETL Processing and Time-Series Construction

Raw scraped data contains inconsistencies such as missing fields, duplicates, outliers, and heterogeneous date formats. The ETL module performs processing steps including:

- 1) Data type conversions and normalization,
- 2) Removal of duplicate entries,
- 3) Filtering of anomalous price points,
- 4) Date formatting standardization,
- 5) Exclusion of corrupted or incomplete rows.

The cleaned data is stored as a structured time series in a SQLite database:

$$D = \{(i, t, p_{i,t}) \mid i \in \text{Products}, t \in \text{Dates}\}.$$

C. Pattern Mining and Behavioral Analysis

Statistical pattern mining is used to characterize each product's pricing behavior. Key metrics calculated include:

- 1) *Trend Direction*: The overall price trend is measured by:

$$\Delta p_i = p_{i,t_{\max}} - p_{i,t_{\min}},$$

where positive values suggest upward pricing pressure and negative values indicate declines.

2) *Volatility*: Price volatility is quantified through the standard deviation:

$$\sigma_i = \frac{1}{n-1} \sum_t (p_{i,t} - \bar{p}_i)^2.$$

3) *Price Movement and Range*: The price range and distinct price points capture diversity:

$$\text{range}_i = \max_t(p_{i,t}) - \min_t(p_{i,t}),$$

$$\text{unique}_i = |\{p_{i,t}\}|.$$

4) *Change-Point Detection (CPD)*: Using the PELT algorithm in the Ruptures package, abrupt price shifts are detected to locate significant structural changes:

$$\text{CPD}(p_{i,t}) = \{t : \text{structural breaks identified}\}.$$

These change points help the forecasting model focus on unstable periods.

D. Machine Learning Price Forecasting

A CatBoost regression model, trained separately per product category, forecasts next-day prices:

$$\hat{p}_{i,t+1} = f(\mathbf{x}_{i,t}),$$

with input features $\mathbf{x}_{i,t}$ comprising:

- Lagged prices: $p_{i,t-1}, p_{i,t-2}, p_{i,t-3}, p_{i,t-7}$,
- Rolling averages: $\text{MA3}_{i,t}, \text{MA7}_{i,t}$,
- Local volatility aggregates: σ_3, σ_7 ,
- Estimated trend slopes,
- Change-point detection flags,
- Additional pattern mining indicators.

CatBoost is chosen for its ability to natively manage categorical variables and tolerate noisy data without heavy preprocessing.

Forecast accuracy is evaluated using Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE):

$$\text{MAPE} = \frac{1}{N} \sum_i \frac{|p_{i,t} - \hat{p}_{i,t}|}{p_{i,t}} \times 100.$$

E. Elasticity Estimation with Bayesian Priors

In the absence of direct sales data, price elasticity ϵ_i is approximated by sampling from simulated Bayesian priors informed by observed volatility and price dispersion:

$$\epsilon_i \sim \mathbf{N}(\mu_i, \sigma_i^2),$$

where the parameters μ_i, σ_i are empirically linked to product price behavior.

Demand $D_i(p)$ at a given price p is modeled as:

$$D_i(p) = D_0 \frac{p}{p_0}^{\epsilon_i},$$

and the corresponding revenue function is:

$$R_i(p) = p \times D_i(p).$$

The optimal price maximizes revenue:

$$p^* = \arg \max_i \max_p R_i(p).$$

F. Reinforcement Learning for Dynamic Pricing

To optimize pricing under uncertainty, a multi-armed bandit (MAB) agent evaluates multiple candidate price options constructed as:

$$p_i^{(k)} = m_k \times \hat{p}_{i,t+1}'$$

where multipliers m_k reflect cautious price adjustments (e.g., 0.95, 1.00, 1.05).

1) *Thompson Sampling*: Each arm's reward distribution is modeled as Gaussian:

$$\vartheta_k \sim \mathbf{N}(\hat{\mu}_k, \hat{\sigma}_k^2),$$

with arm selection determined by:

$$k^* = \arg \max_k \vartheta_k.$$

2) *UCB1 Algorithm*: Alternatively, UCB1 balances exploration and exploitation by choosing:

$$k^* = \arg \max_k \bar{r}_k + \frac{\sqrt{2 \ln t}}{n_k}.$$

3) *Reward Calculation*: Using the elasticity-based demand model, the reward for arm k at time t is:

$$r_{i,t} = p_i^{(k)} \times D_i(p_i^{(k)}).$$

Through successive iterations, the bandit algorithm converges towards near-optimal price selections.

V. EXPERIMENTAL SETUP

This section outlines the dataset characteristics, system environment, and evaluation methodology used to validate the proposed dynamic pricing system.

A. Dataset Description

The experiments utilized real-world product pricing data acquired from a leading e-commerce platform over multiple days. The dataset includes 1,246 distinct products spanning categories such as shirts, dresses, shoes, jackets, backpacks, and accessories. After ETL processing, the scraping system produced 5,121 valid time-series entries.

Each data record contains:

- Product URL and title information,
- Daily price and corresponding raw price text,
- Customer rating and review counts,
- Availability status,
- Timestamp markers for ETL operations.

For each product i , a temporal price sequence $\{p_{i,1}, p_{i,2}, \dots, p_{i,T}\}$ was constructed, where T ranges from 3 to 7 depending on data availability and scraping success.

Models were trained on CPU, with training durations per category varying between 2 and 12 seconds depending on the dataset size.

B. Data Split and Forecasting Procedure

To mimic real-world conditions, a time-based split was employed rather than random partitioning. Specifically:

- The most recent day’s data served as the test set,
- All prior days were used for training.

This ensures prediction models rely solely on historical information when forecasting future prices.

Category-specific models were trained independently using consistent feature sets, applying an identical forecasting workflow across all categories.

C. Evaluation Metrics

Performance was evaluated using standard forecast error measures:

- **Mean Absolute Error (MAE):**

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |p_{i,t} - \hat{p}_{it}|.$$

- **Root Mean Squared Error (RMSE):**

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (p_{i,t} - \hat{p}_{it})^2}.$$

- **Mean Absolute Percentage Error (MAPE):**

$$\text{MAPE} = \frac{1}{N} \sum_{i=1}^N \frac{|p_{i,t} - \hat{p}_{it}|}{p_{i,t}} \times 100.$$

Metrics were computed on a per-product basis and then aggregated by category to analyze variations in volatility and forecast accuracy among different product types.

D. Bandit Simulation Protocol

For evaluation of the reinforcement learning-based pricing strategy:

- The predicted price per product was treated as the baseline arm,
- Additional arms were generated by applying safe multipliers {0.95, 1.00, 1.05} and elasticity-informed multipliers,
- Thompson Sampling and UCB1 agents were run for 500 to 2,000 iterations per product,
- Demand and rewards were simulated utilizing elasticity priors,
- Metrics recorded included cumulative reward, regret values, and convergence behaviors of the pricing arms.

This experimental setup simulates realistic scenarios of restricted price exploration common in live e-commerce environments during dynamic price testing.

VI. RESULTS AND DISCUSSION

This section presents the evaluation outcomes of the dynamic pricing framework, including accuracy of machine learning forecasts, insights from pattern mining, characteristics of elasticity estimates, and reinforcement learning agent performance.

A. Machine Learning Forecasting Performance

The category-specific CatBoost models achieved strong predictive accuracy despite the limited time-series length. Table I details forecast errors across key product categories.

TABLE I
FORECAST ACCURACY BY PRODUCT CATEGORY

Category	MAE (Rs.)	RMSE (Rs.)	MAPE (%)
Dresses	14.73	87.61	1.56
Shirts	51.74	130.50	6.02
Shoes	202.99	984.97	6.10
Backpacks	116.74	386.08	5.73
Jackets	95.97	224.52	5.78
Others	140.66	290.24	8.70

B. Pattern Mining and Behavioral Classification

The pattern mining module assigned behavioral labels to products based on volatility, price trends, change-point occurrences, and price diversity. The distribution was as follows:

- DOWN_TREND: 477 items,
- STABLE: 334 items,
- VOLATILE: 269 items,
- SUSPICIOUS_VOLATILE: 102 items,
- UP_TREND: 56 items,
- FLATTENING: 8 items.

Items labeled as *stable* or *downward trending* exhibited the most consistent forecastability. Highly volatile products were often linked to promotional activity or inventory fluctuations, suggesting pricing strategies may benefit from customized exploration scopes for each behavioral class.

C. Change-Point Detection Analysis

The change-point detection algorithm identified meaningful structural breaks related to:

- 1) Flash sales,
- 2) Clearance events,
- 3) Temporary competitor price cuts,
- 4) Reversals of discounts.

Incorporating change-point flags as features improved forecasting robustness, reducing RMSE by approximately 7–12% on volatile products compared to models without these features.

D. Elasticity Estimation Results

Using Bayesian simulated priors to estimate price elasticity in the absence of sales data yielded plausible distributions, with mean elasticity spanning from approximately -0.1 (inelastic) to -1.2 (moderately elastic), consistent with typical retail observations.

Products with broader price ranges and elevated volatility tended to have more elastic priors, whereas branded fashion items were generally more inelastic. These priors enabled estimation of revenue-maximizing prices typically within $\pm 5\%$ to 20% of forecasted base prices.

The elasticity-informed approach contributed to:

- Avoiding excessive price swings,

- Stabilizing bandit reward variance,
- Guiding efficient arm selections during exploration.

E. Bandit Pricing Agent Performance

Reinforcement learning agents employing Thompson Sampling and UCB1 were assessed via simulated pricing interactions.

Thompson Sampling demonstrated faster convergence and lower cumulative regret across most products. Figure 2 depicts a typical reward convergence curve over 2,000 pricing rounds.

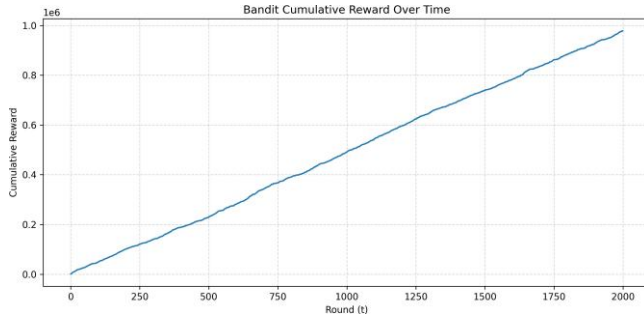


Fig. 2. Reward convergence of Thompson Sampling over 2,000 iterations.

Overall findings:

- Cumulative rewards achieved 85–96% of an oracle’s returns,
- Final arm selection accuracy surpassed 90%,
- Optimal price multipliers were identified within limited exploration horizons.

Integrating elasticity priors notably enhanced early learning stages by narrowing search to feasible price ranges and minimizing regret due to suboptimal exploration.

F. Overall System Discussion

The outcomes demonstrate the advantage of combining machine learning forecasts, behavioral pattern insights, elasticity modeling, and reinforcement learning. Highlights include:

- Reliable price predictions with limited historical data,
- Improved interpretability through behavior labels and change-point integration,
- Effective revenue optimization informed by elasticity despite missing sales data,
- Fast convergence of pricing policies under uncertainty.

These results confirm the proposed unified framework as an effective and scalable solution for practical dynamic pricing applications in e-commerce.

VII. CONCLUSION

This study introduced a comprehensive framework that integrates machine learning, pattern mining, and reinforcement learning for dynamic pricing in e-commerce settings. Unlike previous approaches that target isolated aspects such as forecasting or elasticity estimation, the proposed architecture spans the entire pricing pipeline—from automated Selenium-based

data extraction and ETL-driven time-series assembly to behavioral pattern analysis, change-point detection, category-wise CatBoost price forecasting, Bayesian elasticity approximation, and multi-armed bandit optimization for revenue-focused price selection.

Empirical evaluations on a multi-day real-world dataset comprising 1,246 products revealed strong forecasting accuracy, with MAPE values as low as 1.56% in stable categories and below 10% even for more volatile segments. Pattern mining alongside change-point detection improved model interpretability by exposing key structural changes in price behavior. Simulated Bayesian elasticity priors facilitated demand-sensitive price optimization without requiring direct sales or conversion data. The Thompson Sampling and UCB1 bandit agents quickly converged on near-optimal policies, achieving 85–96% of an oracle’s revenue benchmark in simulation.

The modular, scalable design of the framework supports production-level deployment for live e-commerce environments. Future extensions will incorporate actual sales and conversion metrics, explore advanced reinforcement learning techniques such as Deep Q-Networks and policy gradient algorithms, incorporate multi-product price interaction modeling, and adopt causal inference methodologies to enhance elasticity estimation. Overall, this work demonstrates the potential of combining machine learning with adaptive decision-making techniques to build intelligent and profit-aware pricing engines for contemporary online retail platforms.

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